

Interacting Multipoles of Rank 4 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (1)$$

Simplify deltas

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (2a)$$

$$\xrightarrow{\text{cleaned up:}} R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (2b)$$

Simplify R

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (3a)$$

$$\xrightarrow{\text{cleaned up:}} R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (3b)$$

Simplify Multipoles

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (4a)$$

$$\xrightarrow{\text{cleaned up:}} R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (4b)$$

Exploit Traceless Conditions

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (5a)$$

$$\xrightarrow{\text{cleaned up:}} R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (5b)$$

Reduce Indices

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (6a)$$

$$\xrightarrow{\text{cleaned up:}} R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (6b)$$

Add up

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot -\frac{1}{135135} \cdot [+0] \quad (7)$$

Combining everything

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = [+0] \tag{8}$$